and observe that $AB \neq BA$, since

$$AB = \begin{bmatrix} 5 & 0 \\ 4 & -2 \end{bmatrix} \text{ and } BA = \begin{bmatrix} 0 & 2 \\ 5 & 3 \end{bmatrix}$$

In this case we say that A and B do not commute (with respect to multiplication)

Observe that $AB = BA$ is also possible. For instance,

$$\text{if } A = \begin{bmatrix} 2 & -2 \\ -2 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix} \text{ then, } AB = BA = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}$$

In this case we say that A and B commute with respect to multiplication.

(ii) Cancellation property does not hold for matrix multiplication. That is, $A \neq O$, B, and C are three square matrices of same order $n \times n$ with $n > 1$, then $AB = AC$ does not imply $B = C$ and $BA = CA$ does not imply $B = C$.

As a simple demonstration of these facts, we observe that for instance,

$$\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

but $$\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \neq \begin{bmatrix} 0 & 0 \\ 2 & 3 \end{bmatrix}.$$

(iii) It is possible that $AB = O$ with $A \neq O$ and $B \neq O$; Equivalently, $AB = O$ is not necessarily imply either $A = O$ or $B = O$. The following relation demonstrates this fact :

$$\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

(3) In general, for any two matrices A and B which are conformable for addition and multiplication, for the below operations, we have

- $(A \pm B)^2$ need not be equal to $A^2 \pm 2AB + B^2$
- $A^2 - B^2$ need not be equal to $(A + B)(A - B)$.

Example 7.10

If $A = \begin{bmatrix} 1 & -1 & 2 \\ -2 & 1 & 3 \\ 0 & -3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -3 \\ -1 & 1 \\ 1 & 2 \end{bmatrix}$ find AB and BA if they exist.

Solution

The order of A is 3×3 and the order of B is 3×2. Therefore the order of AB is 3×2.

A and B are conformable for the product AB. Call $C = AB$. Then,

c_{11} = (first row of A) (first column of B)

$$\Rightarrow \; c_{11} = \begin{bmatrix} 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = 1 + 1 + 2 = 4, \text{ since } c_{11} \text{ is an element.}$$

Similarly $c_{12} = 0, c_{21} = 0, c_{22} = 13, c_{31} = 7, c_{32} = 5.$

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